Maz'ya's Critical Uniform-Truncation Conjecture Is False

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Status and the exact 2008 conjecture

This is a `literature_implied_answer (full negative)` packet. Maz'ya's Theorem 1 [1] proves a subcritical estimate for the Newtonian potential $u = \Gamma * f$, where $f \in L^1(\mathbb{R}^n)$ has mean zero. Immediately afterward, on PDF page 2, he conjectures that at $p = n/(n-1)$ one still has

$$\sup_{R>0} \left| \int_{|x|<R} \Phi(\nabla u(x))\, dx \right| \leq C \|f\|_1^p \tag{M}$$

whenever Φ is positively p-homogeneous, Lipschitz on the sphere, and

$$\int_{S^{n-1}} \Phi(\theta)\, d\sigma(\theta) = 0. \tag{S}$$

The weight in the source's formula is one at the critical exponent.

2 Inequality for scalar functions

Theorem 1. *Let f and Φ denote scalar real-valued functions defined on $\mathbb{R}^n$. Assume that $f \in L^1$ and*

$$\int f(x)dx = 0. \tag{4}$$

Furthermore, let Φ be Lipschitz on the unit sphere S^{n-1} and positively homogeneous of degree $q \in [1, \frac{n}{n-1})$. By u we mean the Newtonian (logarithmic for $n = 2$) potential of f:

$$u(x) = \int \Gamma(x-y)\, f(y)\, dy,$$

where $\Gamma(x)$ is the fundamental solution of $-\Delta$.

A necessary and sufficient condition for the inequality

$$\sup_{R>0}\Big| \int_{|x|<R} \Phi\big(\nabla u(x)\big)\, |x|^{n(q-1)-q} dx \Big| \le C\Big(\int |f(x)| dx\Big)^q \tag{5}$$

to hold for all f is

$$\int_{S^{n-1}} \Phi(x)\, d\omega_x = 0. \tag{6}$$

The constant C in (5) depends only on Φ, q, and n.

Here and elsewhere $d\omega_x$ is the area element of the unit sphere S^{n-1} at the point $x/|x|$.

Conjecture. It seems plausible that the inequality (5) holds also for the critical value $q = n/(n-1)$. The following simple assertion obtained in [MS] speaks in favour. The inequality

$$\Big|\int_{\mathbb{R}^2} \sum_{i,j=1}^{2} a_{ij} \frac{\partial u}{\partial x_i}\frac{\partial u}{\partial x_j}\, dx\Big| \le C\Big(\int_{\mathbb{R}^2} |\Delta u|\, dx\Big)^2$$

with $a_{ij} = const$ holds for all $u \in C_0^\infty$ if and only if $a_{11} + a_{22} = 0$.

Proof of Theorem 1. The necessity of (6) can be derived by putting a sequence of radial mollifications of the Dirac function in place of f in (5).

Let us prove the sufficiency of (6). We write $\nabla u(x)$ in the form

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Stolyarov later proved the corresponding *whole-space* inequality [2]. His subsequent bounded-domain theorem [3] shows that a stronger hemisphere cancellation is necessary at a boundary. Applying its necessity construction to a ball, and retaining the zero-mean correction used in the same paper's half-space proof, gives an explicit counterexample to (M).

Full negative answer

Let Γ be the fundamental solution of $-\Delta$ and write

$$K(x) = \nabla\Gamma(x) = c_n \frac{x}{|x|^n}, \qquad c_n \neq 0.$$

Idea of the proof. A ball cuts a concentrated fundamental-solution profile into a hemisphere at its boundary. Choose an odd Φ with nonzero hemisphere mean, concentrate unit mass there, and subtract fixed unit mass to enforce $\int f = 0$. The first piece grows like $\log m$; weak endpoint control makes the correction $O(1)$.

Theorem 1. *For every $n \ge 2$, put $p = n/(n-1)$ and*

$$\Phi(v) = |v|^{p-1} v_1.$$

Then Φ satisfies every hypothesis and the spherical cancellation (S), but there are $f_m \in C_c^\infty(\mathbb{R}^n)$ such that

$$\int f_m = 0, \qquad \|f_m\|_1 \leq 2, \qquad \left| \int_{B(0,1)} \Phi(K * f_m(x))\, dx \right| \longrightarrow \infty.$$

Consequently (M), and hence the source conjecture, is false in every dimension $n \geq 2$.

Proof. The function Φ is positively p-homogeneous and smooth on the unit sphere. It is odd, so (S) holds. Fix nonnegative $\rho, g \in C_c^\infty(\mathbb{R}^n)$ of integral one, with $\operatorname{supp}\rho \subset B(0,1/2)$ and $\operatorname{supp} g \subset B(0,1/4)$. At the boundary point $z = e_1$ of $B(0,1)$ the inward normal is $\xi = -e_1$. Set

$$z_m = (1 - 2/m)e_1, \qquad \rho_m(x) = m^n \rho\big(m(x - z_m)\big), \qquad f_m = \rho_m - g.$$

For all large m, ρ_m is supported in the unit ball; moreover $\int f_m = 0$ and $\|f_m\|_1 \leq 2$.

First omit the compensator and write $A_m = K * \rho_m$. Critical scaling gives

$$\int_{B(0,1)} \Phi(A_m(x))\, dx = \int_{m(B(0,1)-z_m)} \Phi(K * \rho(y))\, dy. \tag{1}$$

The asymptotic calculation in Section 2 of [3] is

$$K * \rho(y) = K(y) + O(|y|^{-n}), \qquad \Phi(K * \rho(y)) = \Phi(K(y)) + O(|y|^{-n-1}). \tag{2}$$

The error in (2) is integrable at infinity. On the sphere,

$$\Phi(K(\theta)) = c_n |c_n|^{p-1} \theta_1,$$

and therefore the inward-hemisphere integral

$$H := \int_{\theta\cdot\xi>0} \Phi(K(\theta))\, d\sigma(\theta) = c_n|c_n|^{p-1} \int_{\theta_1<0} \theta_1\, d\sigma(\theta) \neq 0. \tag{3}$$

For completeness, the logarithmic part of the necessity argument is as follows. Put

$$M = \int_{S^{n-1}} |\Phi(K(\theta))|\, d\sigma(\theta).$$

On radii $4 < r < m^\gamma$, the rescaled ball in (1) converges to the inward half-space, and polar integration of the $(-n)$-homogeneous function $\Phi \circ K$ gives

$$H\gamma \log m + O(1).$$

The remaining radii contribute at most $M(1-\gamma)\log m + O(1)$ in absolute value. Choosing $\gamma < 1$ sufficiently close to one so that $|H|\gamma > M(1-\gamma)$, and using (2), proves

$$\left| \int_{B(0,1)} \Phi(A_m) \right| \longrightarrow \infty. \tag{4}$$

This is precisely the boundary-delta argument proving necessity of the hemisphere condition in [3, Section 2].

It remains to verify that imposing the source's mean-zero constraint does not remove (4). Put $G = K * g$, which is bounded on $B(0,1)$. Positive homogeneity and spherical Lipschitz continuity give the standard estimate

$$|\Phi(a-b) - \Phi(a)| \leq C_\Phi\big(|a|^{p-1}|b| + |b|^p\big). \tag{5}$$

The endpoint weak Hardy–Littlewood–Sobolev estimate yields $\|A_m\|_{L^{p,\infty}} \le C\|\rho_m\|_1 = C$, uniformly in m. Since $p-1<p$ and the ball has finite measure,

$$\sup_m \int_{B(0,1)} |A_m|^{p-1} < \infty.$$

Integrating (5), with $a = A_m$ and $b = G$, therefore shows that

$$\left| \int_{B(0,1)} \Phi(K * f_m) - \int_{B(0,1)} \Phi(A_m) \right| = O(1). \tag{6}$$

The divergence (4) survives in (6). Since the supremum in (M) includes $R=1$, the theorem follows. $\square$

What the later literature does and does not say

The whole-space result of [2] is not contradicted. It controls $\int_{\mathbb{R}^n} \Phi(K*f)$ under full-sphere cancellation, whereas (M) asks for uniform control of every centered-ball truncation. Nonlinear integrals have no domain monotonicity, a distinction emphasized explicitly in [3].

Theorem 1.2 of [3] says that on a bounded smooth domain the correct condition is the pair of hemisphere cancellations

$$\int_{\theta\cdot\xi>0} \Phi(K(\theta))\, d\sigma = 0, \qquad \int_{\theta\cdot\xi>0} \Phi(-K(\theta))\, d\sigma = 0 \quad (\xi \in S^{n-1}).$$

Condition (S) only cancels opposite hemispheres together. The explicit odd Φ above makes that gap visible: its full-sphere mean vanishes, while each relevant hemisphere mean does not.

The supporting paper does not state "Maz'ya's 2008 supremum conjecture is false." It restates the conjecture in the later whole-space form and then studies domains. The identification of its boundary necessity construction with the original uniform-truncation statement, including the zero-mean compensator, is the literature implication established in this packet.

Verification and review recommendation

The source statement was checked in arXiv:0808.0414, PDF page 2. The whole-space theorem and the bounded-domain/half-space theorems were checked in arXiv:2109.08014 and arXiv:2407.14052. The critical scaling identity, hemisphere sign, integrable asymptotic error, and uniform bound for the mean-zero correction were independently audited. Human review is recommended, focusing on the last compensation estimate (6). Confidence in the full negative answer is high.

References

[1] V. Maz'ya, *Estimates for differential operators of vector analysis involving L^1-norm*, arXiv:0808.0414 (2008), Theorem 1 and the conjecture on PDF page 2, arXiv:0808.0414.

[2] D. Stolyarov, *Fractional integration of summable functions: Maz'ya's Φ-inequalities*, arXiv:2109.08014 (2021), Theorem 1.1, arXiv:2109.08014.

[3] D. Stolyarov, *Maz'ya's Φ-inequalities on domains*, arXiv:2407.14052 (2024), Theorems 1.2–1.3 and Section 2, arXiv:2407.14052.